

WINDOWS AUTHENTICATION & ENDPOINT INVESTIGATION

Project 1 — SOC Analyst Portfolio Case Study

Final documented version with evidence screenshots

Component	Details
Environment	Windows 11 SOC laboratory VM
Virtualisation	Oracle VirtualBox
Tools	PowerShell • Windows Event Viewer • Sysmon
Primary telemetry	Security Events 4625 / 4624 • Sysmon Event ID 1
Investigation type	Controlled authentication and endpoint telemetry exercise
Final disposition	Benign / Controlled Lab Activity

Evidence is presented as observed in the controlled laboratory environment.

1. Executive Summary

This project demonstrates a controlled investigation of Windows authentication and endpoint activity. Four failed authentication attempts against the local LabUser account were recorded at 01:36:18 on 31 August 2026. A successful authentication for the same account was recorded 13 seconds later at 01:36:31.

The authentication originated from the local host using the IPv6 loopback address ::1. Endpoint telemetry was then examined using Sysmon Event ID 1 to correlate process execution with the successful logon and to investigate subsequent command execution.

The activity was intentionally generated in an isolated training environment. Final disposition: Benign / Controlled Lab Activity.

2. Investigation Objectives

- Generate and investigate controlled failed and successful Windows authentication events.
- Analyse Security Event ID 4625 and Event ID 4624.
- Distinguish the account being targeted from the account shown in the Subject fields.
- Assess source address, logon type, authentication package and logon process.
- Correlate authentication with endpoint process creation using Logon ID.
- Analyse parent/child process relationships.
- Assess common system-discovery commands without automatically classifying them as malicious.
- Produce a defensible SOC-style conclusion based on observed evidence and laboratory context.

3. Lab Environment

Component	Configuration
Operating system	Windows 11
Virtualisation	Oracle VirtualBox
Security auditing	Logon/Logoff auditing configured for Success and Failure
Endpoint telemetry	Sysmon installed; Event ID 1 process-creation telemetry available
Accounts	SOCAnalyst2 (analyst context) and LabUser (test account)

4. Investigation Timeline

Time	Event	Assessment
31/08/2026 01:36:18	Security Event 4625 × 4	Four failed authentication attempts against LabUser.
31/08/2026 01:36:31	Security Event 4624	Successful interactive authentication for LabUser.
31/08/2026 01:36:31.958 UTC	Sysmon Event 1	cmd.exe created under LabUser; LogonId 0x1AA763.
Later LabUser session	Sysmon Event 1	whoami, hostname and ipconfig observed under LogonId 0x649075.

5. Failed Authentication — Event ID 4625

The failed authentication event identifies LabUser as the account for which the logon failed. The Subject fields identify SOCAnalyst2 as the account associated with the request. The event was an interactive logon attempt from the local machine.

Field	Observed value
Event ID	4625
Time	31/08/2026 01:36:18
Account for which logon failed	LabUser
Account domain	SOCAnalyst

Subject account	SOCAnalyst2
Logon Type	2 — Interactive
Failure reason	Unknown user name or bad password
Status	0xC000006D
Sub Status	0xC000006A
Workstation	SOCANALYST
Source Network Address	::1
Authentication Package	Negotiate
Logon Process	seclogo

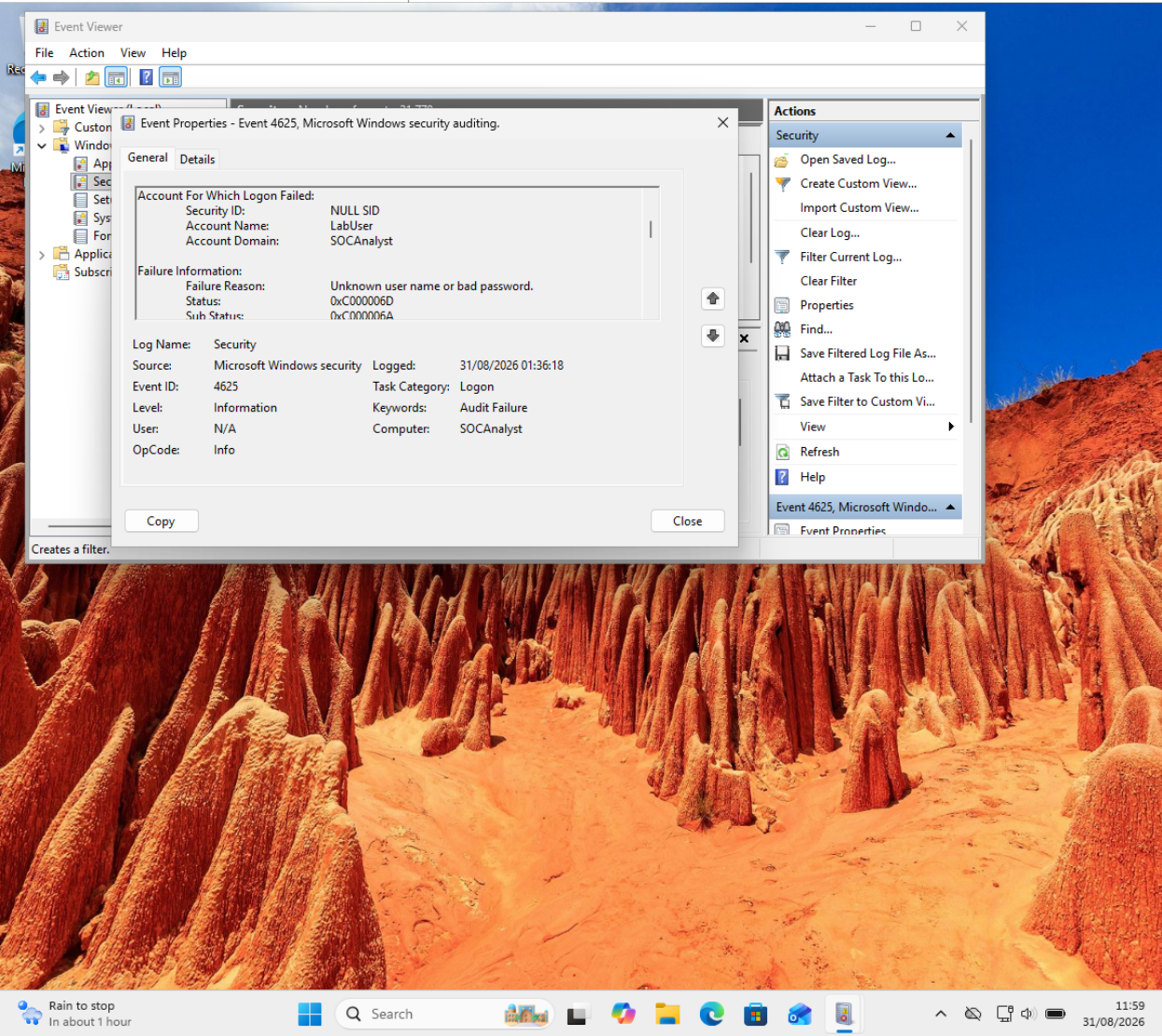


Figure 1 — Windows Security Event ID 4625 showing the failed authentication against LabUser at 31/08/2026 01:36:18.

6. Successful Authentication — Event ID 4624

Thirteen seconds after the failed attempts, Event ID 4624 recorded a successful interactive authentication for LabUser. The successful logon generated Logon ID 0x1AA763, which became the key identifier for subsequent endpoint correlation.

Field	Observed value
Event ID	4624
Time	31/08/2026 01:36:31
New Logon account	SOCAnalyst\LabUser

Logon Type	2 — Interactive
Logon ID	0x1AA763
Workstation	SOCANALYST
Source Network Address	::1
Logon Process	seclogo
Authentication Package	Negotiate
Process ID	0x203c
Process Name	C:\Windows\System32\svchost.exe

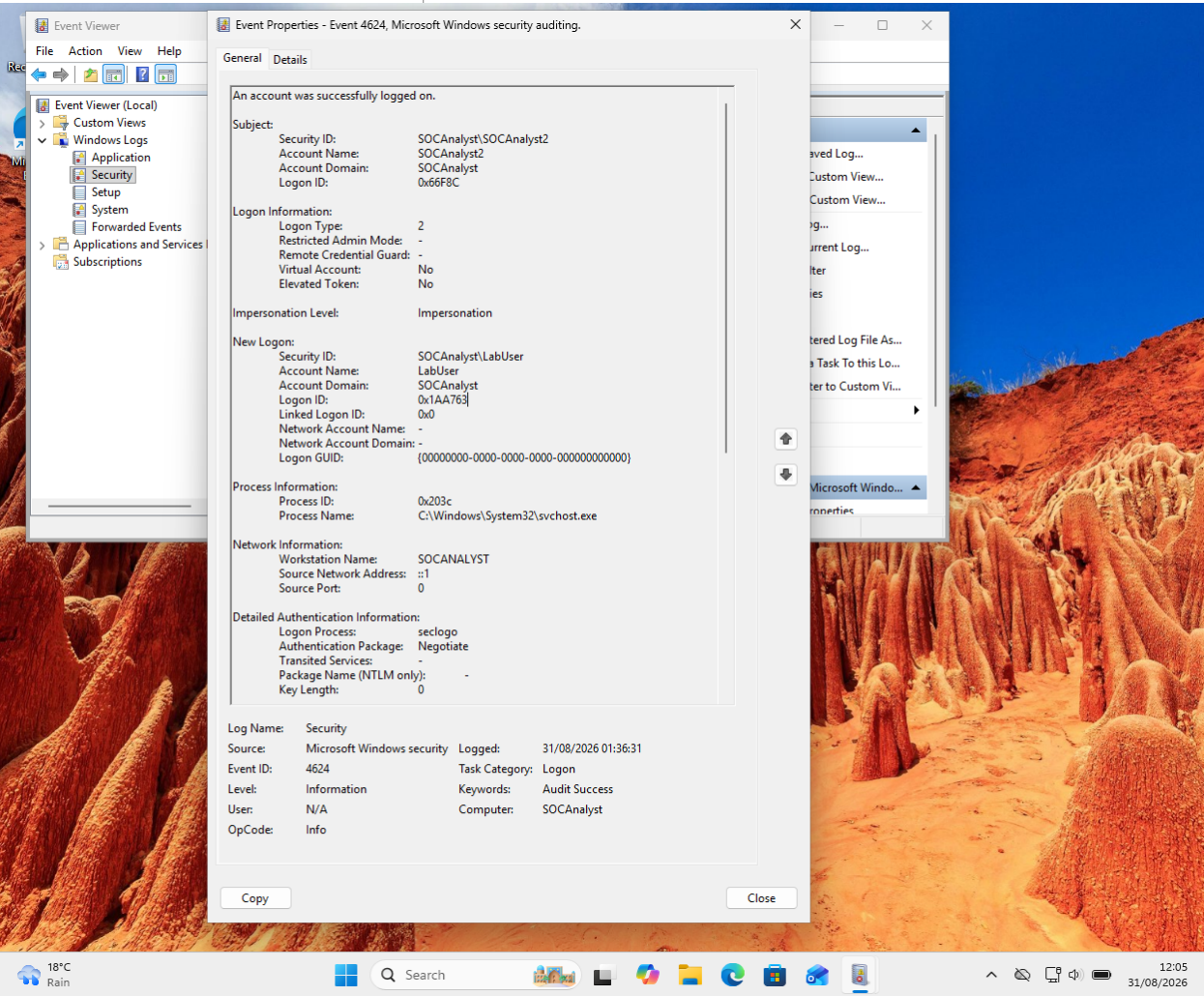


Figure 2 — Windows Security Event ID 4624 showing successful interactive authentication of LabUser. The Logon ID 0x1AA763 is used for endpoint correlation.

7. Sysmon Correlation — cmd.exe

Sysmon Event ID 1 recorded creation of cmd.exe at 01:36:31.958 UTC. The event contained the same Logon ID, 0x1AA763, as the successful 4624 event. This provides strong direct correlation between the successful authentication and the creation of the LabUser command shell.

Field	Observed value
Sysmon Event	1 — Process Create
UTC time	2026-08-31 00:36:31.958
Process ID	8220
Image	C:\Windows\System32\cmd.exe

Command line	"C:\WINDOWS\system32\cmd.exe"
User	SOCAnalyst\LabUser
LogonId	0x1AA763
TerminalSessionId	1
Integrity Level	Medium
Parent Process ID	8468
Parent Image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
Parent User	SOCAnalyst\SOCAnalyst2

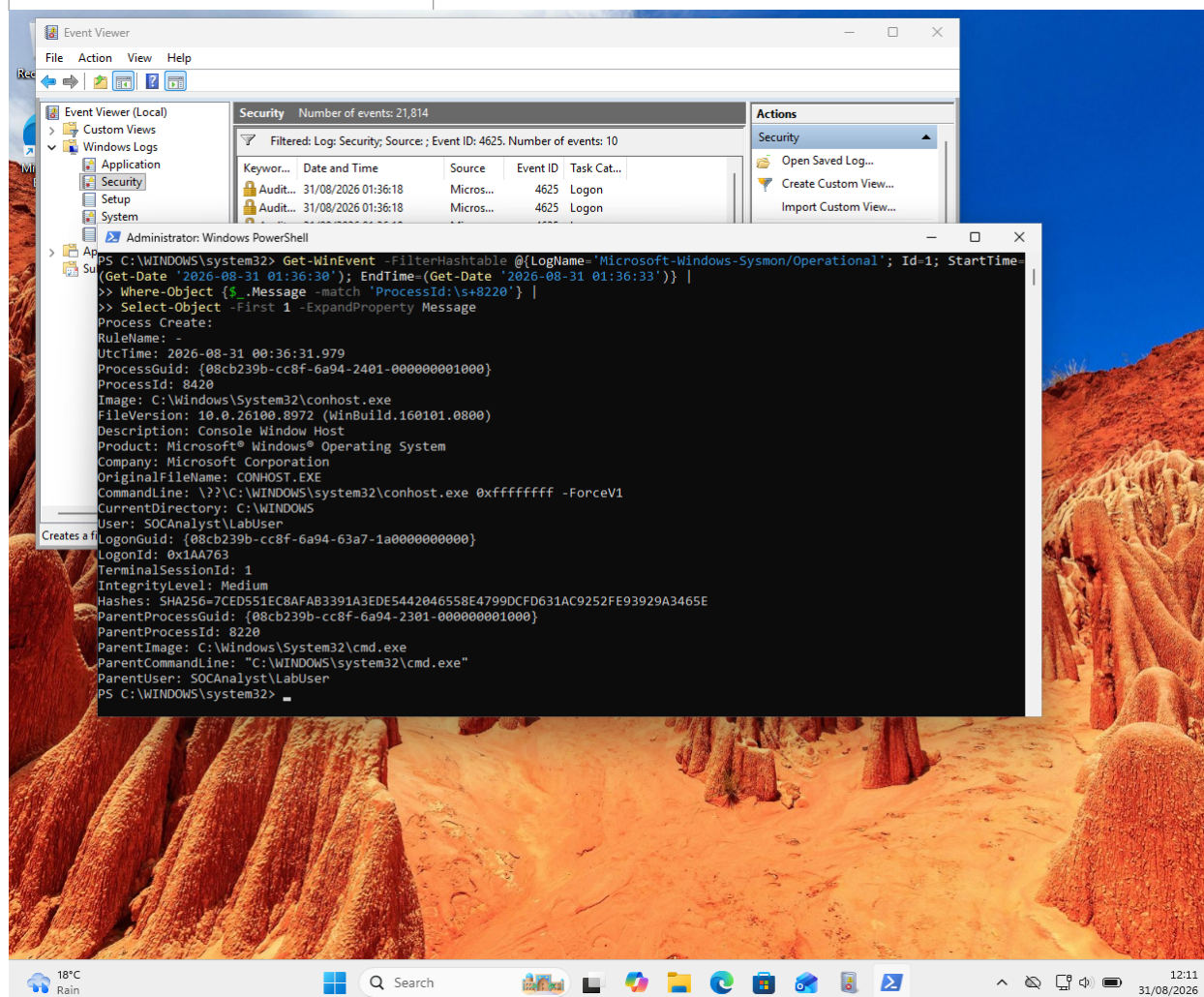


Figure 3 — Sysmon Event ID 1 showing cmd.exe running as LabUser with LogonId 0x1AA763 and a PowerShell parent running as SOCAnalyst2.

Timezone note: the Sysmon timestamp is recorded in UTC, while the Windows Security event is displayed in the VM's local time. The one-hour difference is consistent with the VM's timezone and does not represent a 1-hour investigation delay.

8. Supporting Process Evidence — conhost.exe

A related Sysmon Event ID 1 was also observed for conhost.exe (PID 8420), with cmd.exe (PID 8220) as its parent. This is supporting process-tree evidence rather than the primary authentication correlation.

Field	Observed value
Process ID	8420
Image	C:\Windows\System32\conhost.exe
User	SOCAnalyst\LabUser

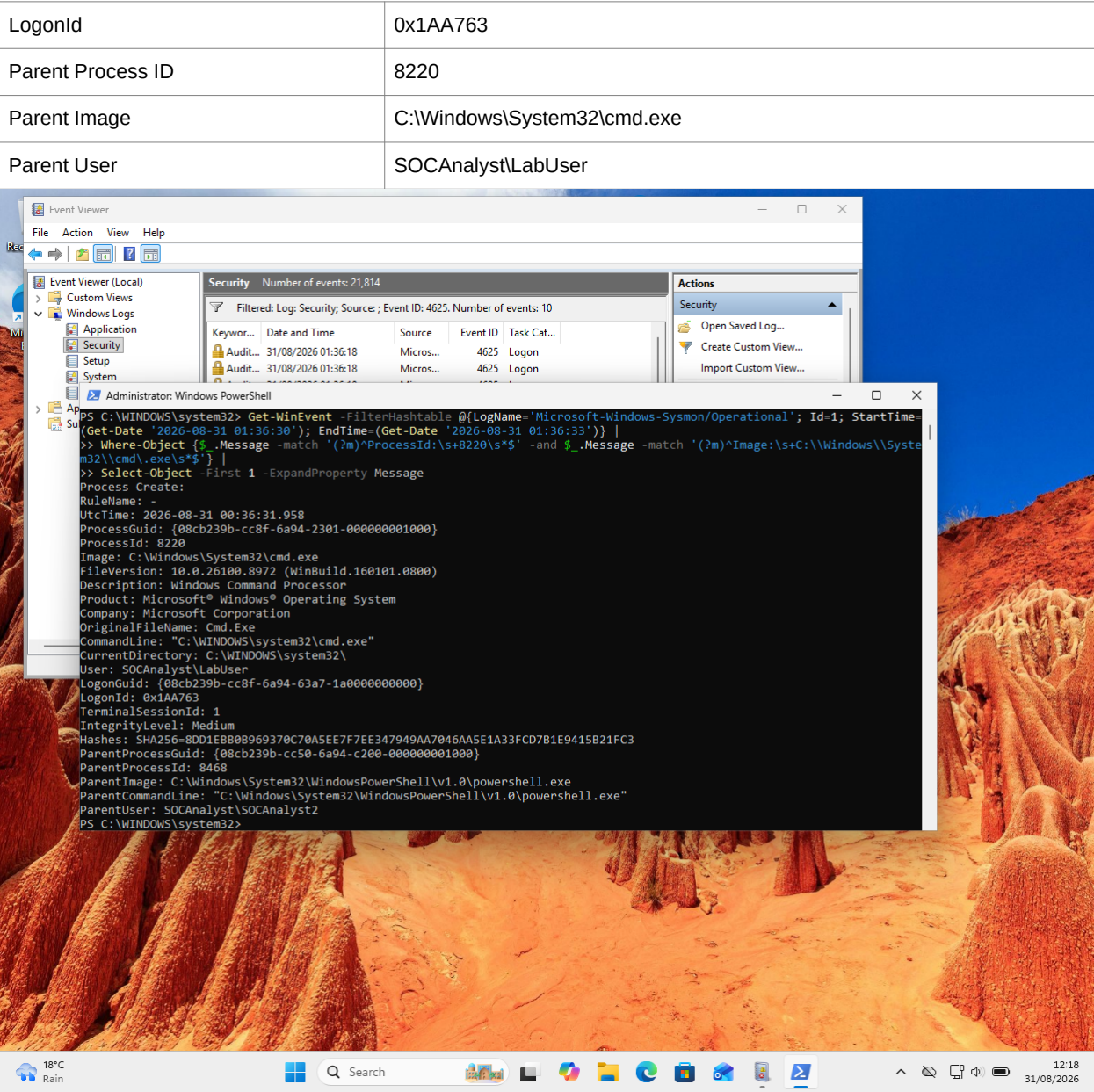


Figure 4 — Supporting Sysmon evidence showing conhost.exe as a child of the correlated LabUser cmd.exe process.

9. Post-Authentication Discovery Activity

A separate LabUser Command Prompt session was used to execute common system-discovery commands.

Process	PID	Command	User	LogonId
whoami.exe	2247	whoami	SOCAnalyst\LabUser	0x649075
HOSTNAME.EXE	1708	hostname	SOCAnalyst\LabUser	0x649075
ipconfig.exe	4132	ipconfig	SOCAnalyst\LabUser	0x649075

These commands are dual-use. They are commonly used by administrators and analysts, but they can also be used during early-stage reconnaissance. Their presence alone does not establish malicious intent.

Evidence-handling note: the discovery commands were associated with LogonId 0x649075, whereas the earlier successful authentication and correlated cmd.exe process used LogonId 0x1AA763. They should therefore be documented as a separate LabUser session unless additional telemetry proves otherwise.

10. Process Relationship

Primary authentication correlation
SOCAnalyst2 → PowerShell (PID 8468) → cmd.exe (PID 8220) → LabUser, LogonId 0x1AA763 → conhost.exe (PID 8420).

Separate discovery session

LabUser → cmd.exe (PID 352) → whoami.exe / hostname.exe / ipconfig.exe, LogonId 0x649075.

11. Analyst Assessment

The initial pattern warranted investigation: four failed authentication attempts against LabUser occurred at the same second, followed 13 seconds later by a successful interactive authentication. A local source address (::1) is reassuring in the laboratory context but does not, by itself, rule out malicious local activity.

The successful logon was strongly correlated to cmd.exe through the shared Logon ID 0x1AA763. The subsequent discovery commands were observed in a separate LabUser session and included whoami, hostname and ipconfig. These commands can represent routine administration or early reconnaissance, so context is required.

The final assessment for this exercise is **Benign / Controlled Lab Activity**. There is insufficient evidence in the collected telemetry to conclude that the activity represented a real compromise, persistence, privilege escalation, credential access or malicious network activity.

This assessment deliberately separates observed evidence from inference. If the same pattern appeared on a production endpoint without an expected explanation, the appropriate response would be to investigate the successful session and subsequent activity further rather than immediately dismissing or confirming compromise.

12. MITRE ATT&CK Technique Mapping

Technique	Name	How represented in the lab
T1059.001	PowerShell	PowerShell was used in the controlled credentialed process workflow.
T1059.003	Windows Command Shell	cmd.exe was created and used to execute commands.
T1033	System Owner/User Discovery	whoami identified the current user/security context.
T1082	System Information Discovery	hostname identified the host.
T1016	System Network Configuration Discovery	ipconfig inspected network configuration.

The technique mappings describe behaviours represented in the laboratory exercise. They are not, by themselves, evidence that the activity was malicious.

13. Key Lessons Learned

- Event ID 4625 represents a failed logon; Event ID 4624 represents a successful logon.
- The Subject account in an authentication event is not necessarily the account being targeted.
- ::1 is the IPv6 loopback address and indicates local origin.
- Logon IDs are valuable correlation pivots across Windows Security and Sysmon telemetry.
- Historical process telemetry is more reliable for investigation than querying a PID that may already have exited.
- Legitimate Windows binaries such as svchost.exe, PowerShell and cmd.exe require contextual analysis.
- System-discovery commands are dual-use and should be assessed alongside timing, user context, parent process and other telemetry.
- A good SOC conclusion distinguishes observation, hypothesis and confirmed evidence.

14. Recommended SOC Follow-Up

- Correlate authentication failures and successes by account, timestamp, source and Logon ID.
- Review process creation immediately following unusual successful authentication.
- Inspect PowerShell command-line telemetry and parent-child process relationships.
- Look for persistence, privilege escalation, credential access and unusual network connections if the activity is not expected.
- Validate whether the user, host, timing and command sequence match normal behaviour.
- Preserve relevant Security and Sysmon events before log retention removes them.

15. Evidence Inventory

Evidence	Purpose	Status
Figure 1 — Security 4625	Failed LabUser authentication	Captured
Figure 2 — Security 4624	Successful LabUser authentication / Logon ID	Captured
Figure 3 — Sysmon cmd.exe	Authentication-to-process correlation	Captured
Figure 4 — Sysmon conhost.exe	Supporting process-tree evidence	Captured
Discovery process data	whoami / hostname / ipconfig in separate session	Documented from collected telemetry

16. Portfolio-Ready Project Summary

Investigated a simulated Windows authentication sequence involving four failed logons followed by a successful interactive authentication. Analysed Security Events 4625 and 4624, identified local authentication via ::1, and used Logon ID 0x1AA763 to correlate the successful logon with Sysmon Event ID 1 process creation. Reconstructed the parent-child relationship between PowerShell and cmd.exe and assessed subsequent system-discovery activity including whoami, hostname and ipconfig. Applied evidence-based reasoning to distinguish suspicious indicators from confirmed malicious behaviour and assessed the exercise as Benign / Controlled Lab Activity.